

Question 3 – Stage 1

You are going to roll five fair six-sided dice with sides labelled 1 to 6.

Your aim is to insert your five scores into the boxes below to create a 5-digit number (i.e. a number in base 10 between 11111 and 66666) that is divisible by 25.

What is the probability (to 3d.p.) that this is possible?

Code

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In [1]: from itertools import product

count = 0
success = 0

for i in product([1,2,3,4,5,6],repeat=5):
    count += 1
    if (2 in i) and (5 in i):
        success += 1

print(f"Successes: {success}")
print(f"Count: {count}")
print(f"Probability: {round(success/count,3)}")

Successes: 2550
Count: 7776
Probability: 0.328
```

Maths

For it to be divisible by 25, it must end in 25, therefore there must be at least one 2 and one 5.

$$p_{12} = \left(\frac{5}{6}\right)^5$$

$$p_{15} = \left(\frac{5}{6}\right)^5$$

$$p_{1(2,5)} = \left(\frac{4}{6}\right)^5$$

$$\begin{aligned} p_{\text{divisible}} &= 1 - p_{12} - p_{15} + p_{1(2,5)} \\ &= 1 - \left(\frac{5}{6}\right)^5 - \left(\frac{5}{6}\right)^5 + \left(\frac{4}{6}\right)^5 \\ &= 0.327932\dots \\ &\approx 0.328 \end{aligned}$$